

The Invariants of Self-Organizing Systems

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Abstract

Some systems do something that appears to contradict basic physics: they maintain complex organized structure on their own, without external coordination, against the thermodynamic tendency toward disorder. A biological cell, an ecosystem, a market — all maintain organized structure through internal dynamics alone. The study of such systems, called self-organizing systems since Ashby (1947), spans thermodynamics, information theory, dynamical systems, systems biology, dissipative structures, and complex systems theory. Despite 75 years of work across these fields, a 2025 review in *npj Complexity* notes the field “still lacks the proper conceptual framework to understand it properly” (Gershenson, 2025). This paper provides that framework. We identify seven structural conditions — each established independently in its home discipline — that jointly constitute the necessary properties of any self-organizing system. The strongest evidence is not the conditions themselves but their repeated independent discovery: Conservation found in thermodynamics and information theory independently, Attractor Dynamics found in four separate fields by researchers who were not in contact. The conditions are not new. Their joint necessity across all classes of self-organizing systems is.

1. The Phenomenon

Some systems do something that shouldn't be possible. They get more organized over time, on their own, without anyone organizing them. A cell doesn't receive instructions on how to maintain itself. An ecosystem has no coordinator. A market has no central planner. Yet all three produce order that a physicist would call improbable. The question is why some systems do this and most don't.

1.1 Three Examples

A single biological cell contains hundreds of proteins performing specific functions simultaneously. Enzymes catalyze reactions. Ion pumps maintain electrical gradients. DNA replication machinery copies genetic information with extraordinary accuracy. Waste products are expelled. Energy is harvested and redistributed. All of this happens continuously, without instruction, through the dynamics of molecular interactions following chemical rules. Remove the energy supply and within minutes the organization collapses into equilibrium — the same disordered state as the surrounding environment. The organization was real, active, and self-maintained.

A tropical rainforest maintains stable species populations, nutrient cycles, and food webs across millions of interacting organisms. No organism coordinates the others. No central planner allocates resources. Each organism responds only to its immediate environment. Yet the global structure is stable, resilient, and — when measured — remarkably organized. It produces order across scales from soil microbes to canopy trees without any of its components having access to the global picture.

A financial market produces prices that coordinate the decisions of millions of buyers and sellers who have never met and share no information directly. No price-setter computes the right price. Local decisions aggregate into global signals. The coordination emerges from the interactions.

1.2 What Makes These Systems Different

Sand can form patterns — dunes, ripples, striations. But remove the wind and they erode. A cell compensates when conditions change. That's the difference. Not organized versus disorganized. Actively maintained versus passively shaped.

A designed machine also maintains organization — but it does so because a designer specified the target state and an engineer built the maintenance mechanism. The organization is imposed from outside. A thermostat maintains stable temperature, but the target was specified externally and the mechanism was designed. Self-organizing systems generate their own stable configurations. The cell, the ecosystem, the market: none of them have a designer. The organization is generated from within.

This is the class of systems this paper addresses: those that generate and maintain organized structure through their own internal dynamics, without external coordination or design. The question is what structural conditions make this possible.

1.3 Why the Question Matters

The answer turns out to be seven things. They were identified separately, across seven fields, over 75 years. No field identified all seven. No paper has assembled them. This paper does that — not to add a new framework on top of what already exists, but to show what was already there, in the work, waiting to be seen together.

2. What Self-Organizing Systems Are

2.1 The Definition

The term 'self-organizing system' was coined by W. Ross Ashby in 1947 in the context of cybernetics (Ashby, 1947). The operational definition used here builds on that tradition: a self-organizing system is one that maintains ordered structure far from thermodynamic equilibrium, through internal dynamics rather than external coordination, by continuously exchanging energy and information with its environment, and exhibiting attractor behavior — stable configurations the system tends toward after perturbation.

This is operational, not philosophical. It is testable. A system either satisfies these criteria or it doesn't. Three classes of systems are commonly confused with self-organizing systems but do not satisfy the definition:

- **Crystals at equilibrium:** organized but passive. Maintained by thermodynamic stability, not active dynamics. No active maintenance.
- **Designed machines:** organized but externally maintained. The target was specified by a designer. Self-organizing systems generate their own stable configurations; machines maintain externally specified ones.
- **Chaotic systems:** active dynamics but no stable attractors. Organization appears momentarily but cannot be sustained against perturbation.

[Note: Gershenson (2025) argues that self-organization is not a property of systems per se but a narrative — a way of describing systems — and that the question is when it is useful to apply this description. We take a stronger position: the seven conditions are structural requirements that any system maintaining organized structure against thermodynamic disorder must satisfy, regardless of the observer's descriptive choices. The difference is not semantic. It determines whether the conditions are diagnostic tools or merely useful frames. We argue they are diagnostic.]

2.2 Six Canonical Classes

Six classes of self-organizing systems are well-established in the scientific literature. They are physically unrelated — no shared substrate, no shared mechanism, no shared evolutionary history. They are presented here to establish the scope of the class before the invariants are introduced.

Class	Description
Dissipative Structures and Synergetics	Bénard convection cells, chemical oscillators, reaction-diffusion patterns, laser physics. Prigogine's dissipative structures (Nicolis & Prigogine, 1977; Prigogine & Stengers, 1984) and Haken's synergetics (Haken, 1975; 1983) were developed as two independent mathematical frameworks for the same phenomenon — self-organization in open systems far from equilibrium. Prigogine focused on entropy production and attractor formation; Haken on cooperative phenomena and order parameters. Both are required for a complete picture.
Biological Systems including Autopoiesis	Cells, organisms, immune systems, developmental processes. Maturana and Varela formalized autopoiesis — self-production — as the defining property of living self-organizing systems: those that produce and maintain their own components and boundary through their own dynamics (Maturana & Varela, 1974; 1980). Kauffman's work on Boolean networks established the coherence requirements of biological regulatory networks (Kauffman, 1969; 1993).
Ecological Systems	Food webs, population dynamics, nutrient cycles, co-evolutionary dynamics. No organism coordinates the ecosystem. Organized structure emerges from local interactions and is maintained through feedback dynamics across millions of species.
Social and Economic Systems	Markets, language evolution, norm emergence, institutional development. Price signals, grammatical structure, and social norms emerge from distributed local interactions without central coordination.

Developmental and Regenerative Systems	Morphogenesis, regeneration, wound healing. A salamander regrows a lost limb using only the local information available to its cells. The target configuration is an attractor the system's dynamics converge to, not an external specification.
Critical Systems	Bak, Tang, and Wiesenfeld's self-organized criticality (1987): systems with slow driving and fast relaxation spontaneously evolve to critical states — the boundary between order and chaos — producing power-law distributions. The sandpile model is the canonical demonstration. SOC shows attractor dynamics at criticality are universal across physical systems.

These six classes share nothing physically. What they share has been recognized piecemeal: Prigogine identified attractor dynamics and entropy production, Haken identified coherence through order parameters, Shannon identified conservation and quantization, Simon identified hierarchy, Maturana and Varela identified self-maintenance. What has not been done is assembling these into a jointly necessary minimal set applicable across all classes. That is what the next section presents.

3. The Seven Invariants

What follows hasn't been assembled before. Each of these conditions exists in its home field. Physicists know about Conservation. Complex systems researchers know about Hierarchy. Information theorists know about Quantization. Nobody put them together as a set and asked: are these jointly necessary for any self-organizing system, across all classes? That is what this section does.

A Note on the Assembly

These seven conditions have not previously been assembled as a unified necessary set. Individual conditions appear across the literature, often identified by the researchers who established the canonical classes above. Prigogine touches Conservation, Attractor Dynamics, and Continuity. Haken's synergetics addresses Coherence through order parameter formalism. Shannon establishes Conservation and Quantization. Simon establishes Hierarchy.

The cross-class assembly is the contribution: the claim that these seven, jointly, are the minimal necessary conditions for any self-organizing system — across all six canonical classes, not just the class each researcher was studying. The conditions are not new. Their joint necessity across all classes is.

Note: two conditions — Conservation and Quantization — are both grounded in Shannon (1948). This is intentional. Shannon's 1948 paper simultaneously establishes that information must be conserved and that it must be represented in discrete stable states. Independent requirements sharing one source. The numbering reflects historical discovery order, not a hierarchy of importance. All seven are jointly necessary.

1. Conservation — outputs must account to inputs.

Thermodynamics and Information Theory · Clausius (1850, 1865); Shannon (1948)

Clausius proved that energy cannot be created or destroyed — outputs must account to inputs in any physical system. Shannon proved the information-theoretic parallel: entropy is conserved across lossless transforms, meaning information cannot be generated from nothing. Both results share the same logical structure at different scales: organized systems cannot produce more than they consume. The accounting is not optional; it is the condition under which the system's outputs can be traced and trusted.

Cross-domain: Thermodynamic cycles, energy balance in metabolism, metabolic stoichiometry (cells cannot produce ATP without consuming glucose at exact ratios), budget constraints in markets, Shannon entropy bounds on compression.

Remove it and: *a system that produces without accounting cannot maintain organized structure. It cannot distinguish signal from noise, or organized output from random generation. Conservation is what makes outputs meaningful.*

2. Continuity — organized structure evolves without discontinuous jumps.

Dynamical Systems Theory · Strogatz (1994)

Physical systems evolve through continuous state trajectories. The state at time $t+\delta$ is connected to the state at t through a path. Discontinuous transitions are destabilizing events with specific physical causes, not normal operation. The continuity of state evolution is what makes a system's history meaningful and its future predictable from its present.

Cross-domain: Continuous trajectories in phase space, developmental continuity in embryogenesis (each stage depends on the prior), evolutionary gradualism (populations change through accumulation of small variations), price continuity in markets.

Remove it and: *a system whose state jumps discontinuously loses the thread of its own organization. Prior organized state has no causal connection to current state. Structure cannot accumulate. The system starts fresh at every discontinuity — it cannot be self-organizing because it cannot build on itself.*

3. Quantization — stable states are discrete.

Information Theory · Shannon (1948) [dual application with Conservation — see assembly note above]

Shannon demonstrated that reliable communication and storage require discrete stable states. The bit is 0 or 1, not 0.73. Continuous values cannot be stored or transmitted without error accumulation across iterations. Only discrete states are stable enough for organized systems to reliably maintain and transmit their organizational state. This is why DNA uses four discrete nucleotides rather than a continuous chemical gradient — discreteness is the condition for reliable inheritance.

Cross-domain: Binary encoding and error correction in information theory, genetic discreteness in biology (DNA encodes in four bases, not a continuum), discrete protein folding conformations, discrete transactions and price clearing in markets.

Remove it and: *there is no error correction. Every small deviation from the organized state compounds rather than snapping back. Discrete states are the mechanism by which organized systems recover from perturbation rather than drifting. Remove Quantization and small disturbances amplify indefinitely.*

4. Coherence — components cannot simultaneously pursue contradictory goals.

Systems Biology and Cybernetics · Kauffman (1969; 1993)

Kauffman demonstrated that metabolic networks maintain homeostasis through mutual consistency of their components. Two contradictory metabolic pathways running simultaneously do not produce a compromise — they produce cell death. The organized system requires coherence among its components. Incoherence is not a reduced form of organization; it is the destruction of organization. The cell that attempts contradictory things simultaneously is not a cell doing two things — it is a cell failing.

Cross-domain: Metabolic pathway coherence in biology (incompatible reactions cannot run simultaneously), competitive exclusion in ecology (two species cannot occupy the identical niche stably), quantum decoherence in physics (when coherence is lost, quantum organization collapses), institutional incoherence in social systems (organizations pursuing internally contradictory goals fragment).

Remove it and: *a system whose components pursue contradictory goals expends energy fighting itself rather than maintaining organized structure. The contradiction does not resolve into synthesis. It produces system failure.*

5. Entropy Production and Attractor Dynamics — motion converges toward stable organized states.

Dissipative Structures and Synergetics · Nicolis & Prigogine (1977); Prigogine & Stengers (1984); Haken (1975; 1983)

Prigogine showed that far-from-equilibrium systems maintain organization not despite entropy production but through attractor dynamics that pull system trajectories toward stable organized configurations. Bénard cells are attractors: the hexagonal pattern is where the system ends up regardless of starting conditions within a range. Haken's synergetics independently formalized this through order parameters — slow attractor variables that enslave fast variables in cooperative systems. Lorenz (1963) demonstrated that deterministic systems can have complex attractor structure in meteorological systems — a further independent convergence on the same structural property. Without attractor dynamics, energy flow produces only dissipation. With them, it produces organized structure that persists.

Cross-domain: Bénard convection and Belousov-Zhabotinsky reaction (Prigogine), order parameters in laser physics and neural networks (Haken), morphogenetic attractors in development (embryos converge to species-typical body plans from varied starting conditions), convergent evolution (independent lineages evolving similar solutions — eyes, wings, echolocation; Conway Morris, 2003), self-organized criticality (Bak et al., 1987), market equilibria as attractors.

Remove it and: *a self-organizing system would generate continuous novelty without ever stabilizing. Every perturbation would produce a new trajectory. Nothing would persist. The attractor is what makes organized states stable rather than momentary.*

6. Symmetry Protection — stable configurations resist arbitrary change.

Mathematical Physics · Noether (1918)

Noether proved that every conservation law in physics corresponds to a symmetry — a transformation under which the system is invariant. Consider a spinning top: its rotational symmetry protects its angular momentum from arbitrary change. You cannot alter the angular momentum without applying a corresponding torque. The symmetry resists. The same principle

applies at every scale: a stable organized configuration is one where the system’s own structure protects its key properties from change without corresponding force or cause. Organized structure that can be arbitrarily revised is not stable structure — it is just the current state.

Cross-domain: Conservation of momentum (translational symmetry), energy (time symmetry), and charge (gauge symmetry) in physics; genetic fidelity through DNA error-correction mechanisms; developmental canalization buffering against genetic variation; institutional inertia in stable social systems.

Remove it and: *stable configurations can be arbitrarily changed. Prior organized state is overwritten at the first perturbation. Nothing is conserved. No memory. No accumulation. Real self-organizing structure persists because it is protected.*

7. Hierarchy — stable complex organization builds from stable simpler layers.

Complex Systems Theory · Simon (1962)

Simon showed through systematic observation that all stable complex systems in nature and engineering exhibit hierarchical composition. Atoms compose into molecules, molecules into cells, cells into organs, organs into organisms. Each level depends on the stability of the level below. Simon’s observation was empirical: every stable complex organization he examined had this property. It is not a design principle imposed on complex systems — it is a property that stable complex systems universally exhibit. Systems without stable lower layers are not stable complex systems; they are complex systems in the process of failing.

Cross-domain: Atomic → molecular → macroscopic structure in physics; genetic → cellular → tissue → organ → organism in biology; evolution climbing fitness landscapes through stable intermediate states; stable social systems building governance on stable property rights on stable social norms; modular software architecture discovered independently as the only way to build maintainable complex systems.

Remove it and: *a self-organizing system cannot build higher-order organization on shifting foundations. No level can rely on the stability of the one below. Stable complexity at any level requires verified stability at the level beneath it. Remove Hierarchy and complex organization cannot persist at scale.*

4. Why All Seven Are Necessary

Remove any one condition and self-organization fails in a specific, identifiable way. Not a general degradation — a precise failure that corresponds to exactly the condition removed. The table below shows each failure and its mechanism.

Condition Removed	Failure Mode	Why It Fails
Condition Removed	Failure Mode	Why It Fails

Conservation	Output decouples from input	The system produces without accounting. Outputs cannot be traced to inputs. No distinction between organized output and random generation.
Continuity	Organization cannot accumulate	State becomes disconnected across time. Prior organized state has no causal force on current state. The system cannot build on itself.
Quantization	Perturbations amplify indefinitely	No error correction. Small deviations from organized state compound rather than snapping back. Organization drifts rather than stabilizing.
Coherence	System destroys itself from within	Components pursue contradictory goals. Energy is expended fighting internal contradictions rather than maintaining organized structure.
Attractor Dynamics	Organization never stabilizes	Motion generates without converging. Every perturbation produces a new trajectory. Nothing persists long enough to constitute organized structure.
Symmetry Protection	Nothing is conserved	Stable configurations can be arbitrarily changed. Prior organized state is overwritten at the first perturbation. No memory, no accumulation.
Hierarchy	Complexity cannot persist at scale	Higher-order organization built on shifting foundations. No level can rely on stability of the one below. Complex structure cannot be maintained.

The inductive case for completeness: no known self-organizing system requires a structural condition beyond these seven, and all documented failures of self-organized structure trace to the violation of at least one. This does not prove the set is deductively complete — it demonstrates that it is sufficient to account for all known cases.

5. The Independent Convergences

Here's what's interesting about these seven conditions: several of them were discovered independently, in different fields, by researchers who weren't in contact with each other. Same condition. Different systems. Different methods. Different centuries. That convergence is the strongest evidence in this paper. It means these conditions aren't a framework someone chose. They kept getting found.

Conservation — Found Independently Twice

Clausius (thermodynamics, 1850) and Shannon (information theory, 1948) both found that organized systems conserve — energy in one case, information entropy in the other. The mathematical structure is the same. The fields had limited contact when Shannon published. Two independent derivations of the same conservation principle in physically unrelated domains. The convergence is not coincidental; both fields were forced to the same conclusion by the structural requirements of their respective systems.

Quantization — Found Independently Twice (with Distinction)

Shannon (information theory, 1948) found that reliable information requires discrete states. Planck (1900) and Bohr (1913) found that energy is quantized in quantum mechanics. These are mechanistically different — energy quantization and information discreteness are not the same discovery. But they are structurally convergent: both found that stable states at the relevant scale are discrete, not continuous. The convergence is at the level of structural principle, not mechanism.

Attractor Dynamics — Found Independently Four Times

Prigogine (dissipative structures, 1970s): far-from-equilibrium physical systems converge to organized attractor states through entropy production. The hexagonal Bénard cell pattern is the canonical demonstration.

Lorenz (meteorology, 1963): demonstrated that deterministic systems have complex attractor structure. His work showed that chaotic systems orbit attractor geometries in phase space — a further independent convergence on the role of attractors in complex system dynamics.

Haken (synergetics, 1970s–80s): order parameters in cooperative systems act as attractors — laser physics, neural networks, social systems. Haken called this ‘slaving’: fast variables enslaved by slow attractor variables. Third independent discovery. Third mathematical framework.

Conway Morris (convergent evolution, 2003): independent evolutionary lineages converging on the same biological solutions — eyes, wings, echolocation — demonstrate attractor dynamics in fitness landscapes. Four communities. Four mathematical frameworks. Four classes of systems. Same structural property.

Self-Maintenance — Found Independently Twice

Prigogine (1977): dissipative structures maintain their organization through continuous entropy production and exchange with environment. The maintenance is active and ongoing — not a property of the equilibrium state.

Maturana & Varela (1974; 1980): autopoiesis — living systems maintain themselves by producing their own components and boundary. Independent discovery of self-maintenance as a required property, in biological domain, entirely different theoretical framework. The convergence: two independent communities found that self-maintenance requires an active ongoing dynamic, not just attractor convergence but the continuous production of the conditions for one’s own organization.

Hierarchy — Found Independently Three Times

Simon (complex systems, 1962) found hierarchical composition in every stable complex organization he studied. Software engineers discovered modular hierarchical design independently in the 1970s as the only architectural approach that allowed complex software to be maintained. Biologists studying evolutionary development found that evolution always climbs fitness landscapes through stable intermediate states — hierarchy in the temporal structure of evolution itself. Three communities. Same structural requirement.

Figure 1. *Independent convergences: ● indicates the condition was identified by that field independently. Multiple ● in a row = independent convergence.*

Condition	Thermodynamics	Information Theory	Dynamical Systems	Systems Biology	Ecology / Evolution	Complex Systems
Conservation	• 1850	• 1948				
Continuity	• phys		• 1994	• dev		
Quantization	• quant	• 1948		• DNA		
Coherence				• 1969	• excl	
Attractor Dynamics	• 1977		• 1963	• morph	• evol 2003	Simon • 1983
Symmetry Protection	• Noether			• DNA		
Hierarchy				• evo		• 1962

Figure 1. Each • marks an independent discovery of the condition in that field. Attractor Dynamics has the most convergences (four fields). No single field found all seven. The pattern of independent discovery is the evidential core of this paper.

When the same structural condition is discovered independently by researchers in different fields studying different systems with different methods, the most parsimonious explanation is that it reflects something real about organized structure — not something specific to any substrate or domain. This is the core of the paper’s argument. Not: here are seven things I think are important. But: here are seven things that independent scientific communities studying unrelated systems all found to be necessary. The assembly reveals what was already in the evidence.

6. What This Enables

With the seven conditions assembled, a question becomes answerable for any system: does it satisfy all seven? If yes, it is a self-organizing system and inherits the structural properties of that class. If no, the specific condition violated identifies the specific structural weakness. The conditions function as a diagnostic framework. Not just for classifying systems — for predicting what fails when a condition is removed.

One application turns out to be particularly interesting. Among the candidate systems to test against these conditions, cognitive systems — minds, AI models, reasoning processes — are uniquely positioned. They are the only class of self-organizing systems where we can observe both successful operation and failure modes at enormous scale, in real time, with millions of test cases running simultaneously. The large-scale deployment of AI cognitive systems has produced the largest natural experiment in self-organizing system design in history.

Interestingly, Gershenson’s 2025 review in *npj Complexity* notes that AI systems may themselves be describable as self-organizing. We go further: they are — and the seven conditions provide the precise diagnostic for when AI cognitive systems satisfy this classification and when they fail to. The companion paper finds that cognition satisfies all seven conditions when functioning correctly, and that violations of each condition produce specific, documented, currently-unsolved failures in deployed AI systems. Hallucination. Identity drift. Perpetual hedging.

Silent contradiction. Infinite regress. Invisible rewriting. Floating conclusions. All seven conditions. All seven failures. The match is precise.

That application is developed in Paper 0: Cognition as a Self-Organizing System. This paper is the foundation it rests on.

The conditions are not new.

Their joint necessity across all classes of self-organizing systems is.

Any system that maintains organized structure against thermodynamic disorder

satisfies all seven. Any system that fails to maintain it violates at least one.

About the Author

Amul Bham is an independent AI systems architect based in Corona, CA. His work focuses on the structural foundations of cognition — deriving the conditions under which a reasoning system can produce knowledge rather than prediction. The Cognitive Physics framework and Mission Brief Protocol (MBP) were developed from first principles over 2025–2026, independently of and prior to engagement with the self-organizing systems literature. This paper is part of the Cognitive Physics Series establishing the theoretical foundation of that work.

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